

Nontraditional Implement Training Methods

Traditionally, resistance training interventions used in most modern strength and conditioning facilities have relied heavily on barbells, dumbbells, and various resistance training machines. More recently, strength and conditioning professionals have begun to incorporate nontraditional implements to potentially improve training specificity and to add greater variation to the preparation of athletes. Nontraditional implements can include items that are typically associated with **strongman training** such as tires, logs, kettlebells, stones, weighted sleds, and other weighted implements (59).

Strongman Training

Strongman implement training has recently seen an increase in popularity as a proposed tool for enhancing sport performance (15, 87, 116, 147, 159, 166). Some of the most common strongman exercises are tire flipping, log (or keg) lifting, farmer's walks, and sled pushes and pulls. While research on these types of exercises is limited, some evidence suggests that they can be used to introduce a high-intensity stimulus resulting in an elevated blood lactate response (16, 73, 166). Additionally, it has been suggested that these types of exercises create a greater degree of instability that effectively challenges the athlete in different ways compared to traditional resistance training activities (87). Recent systematic reviews on the biomechanics of strongman have critiqued the research methodologies employed (63) as well as the results and potential applications of this type of training within strength and conditioning (62).

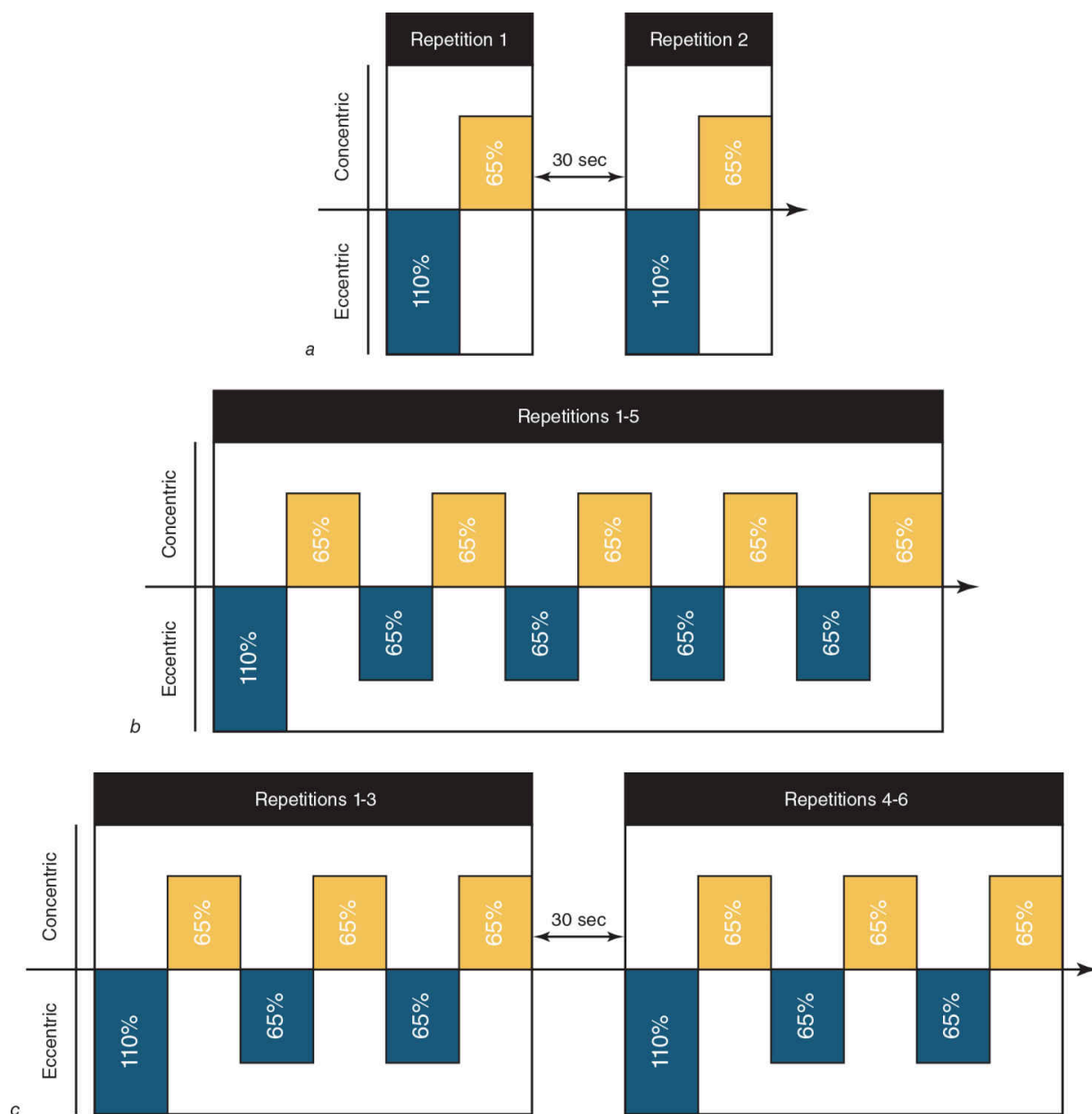


FIGURE 17.8 Example of accentuated eccentric loading (AEL) set structures.

Adapted by permission from M. Harden, P. Comfort, and G.G. Haff, "Eccentric Training: Scientific Background and Practical Applications," in *Advanced Strength and Conditioning: An Evidence-Based Approach*, 2nd ed., edited by A.N. Turner and P. Comfort (Routledge, 2022), permission conveyed through Copyright Clearance Center, Inc.

Research documenting the chronic training-related adaptations from strongman-type training are particularly scarce (3, 60, 96, 157, 162). Winwood and colleagues (157) compared seven weeks of strongman compared to traditional resistance training on body composition, strength, power, and speed measures in 30 resistance-

trained, sub-elite rugby players. While both groups tended to report improvements in strength and functional performance measures (0.2%-7%), no significant differences were observed between groups. This would indicate that strongman training can be equally as effective as traditional training and that its use might provide additional variety and interest to highly experienced resistance-trained athletes who may benefit from more variations in their training stimulus.

Heavy sled pulls have also been the focus of some training studies (96, 157). For example, the training study by Winwood and colleagues (157) attempted to match strongman to traditional resistance training exercises. One example was the use of heavy sled pulls with 85% of the athlete's 1RM back squat for 3 sets of 25 meters (27 yd). Morin and colleagues (96) have also compared eight weeks of heavy sled sprint training (10 × 20 m [22 yd] sprints per session with 80% of body mass) to a control group performing unresisted sprinting on sprinting performance. Heavy sled training produced a range of moderate effect size improvements in horizontal force production and 5-meter (5 yd) sprint performance, with effect sizes indicating that heavy sled training had small to moderate benefits when compared to traditional sprinting. Further, a meta-analysis including 13 studies demonstrated the effectiveness of resisted sled training with a range of loads (5%-80% body mass) for improving acceleration (up to 10 m [11 yd]) but not maximum-velocity sprint performance (3), with no clear indication of what constitutes the most effective loads.

The effect of tire flipping has also been examined in one study (162), whereby 29 physically active males performed 4 sets of 6 repetitions with either lighter (43 kg [95 lb]) or heavier (104 kg [229 lb]) tires for two sessions per week for 8 weeks. Both groups significantly improved 6RM bench press, intermittent endurance, five horizontal jumps and change-of-direction ability, with no significant between-group differences in these responses. Neither group

reported significant improvements in 20-meter (22 yd) sprint times or repeated sprint ability.

Due to the potentially more awkward and unpredictable nature of strongman training compared to traditional resistance training, muscle activation and injury risk may also differ substantially between these training approaches. McGill and colleagues (87) quantified trunk and hip muscle activity and kinetics across the farmer's walk, suitcase carry (unilateral farmer's walk), yoke walk, tire flip, log lift, keg lift, and stone lift. While all of these movements required exceedingly high levels of activation of certain trunk and hip muscles and produced large spinal loads, these biomechanical characteristics exhibited considerable between-exercise variation. Such results suggest that each of these strongman exercises impose somewhat unique challenges on the trunk and hip musculature, suggesting that the inclusion of a variety of strongman exercises may be useful to better develop spinal stability across a range of movement situations.

Regarding injury risk, the somewhat unique challenges imposed by different strongman exercises (87) may be consistent with the views of 213 strongman athletes who felt that certain exercises (tire flip, yoke walk, and stone lift) were more dangerous and likely to contribute to injury risk when compared to other strongman exercises (158). Perhaps as a consequence of the common usage of the tire flip, yoke walk, and stone lift by strongman athletes, two systematic reviews have reported greater injury rates in strongman athletes when compared to other strength sport athletes such as weightlifters, powerlifters, and bodybuilders (74, 121). Consequently, athletes who perform strongman exercises should focus on performing these exercises with appropriate technique and only use loads that allow them to maintain adequate body and implement control.

Tire Flipping

Tire flipping typically employs truck or heavy-equipment tires that can be modified via placement of an extra load in the center to address individual strength needs (59, 147). The selection of the appropriate tire size depends on numerous factors that the strength and conditioning professional must consider when designing a specific program for an athlete. Thought must be given to the tire's dimensions, including the height, width, and weight (21). A key consideration is the tread on the tire; worn treads are more difficult to grip, and pronounced treads potentially contain cuts, debris, or exposed metal that could put the athlete at risk for injury (21, 147).

The most common injury associated with the tire flip is biceps tears (158), which may occur when the hands slip on the tire. To mitigate this injury risk, it is generally recommended that the tires used should be dry and have sufficient tread to allow the athlete to maintain grip and properly execute the exercise (21). Additionally, it is important that the tire is selected based on the individual athlete's strength levels and that the movement is performed with appropriate exercise technique so that injury risk is minimized.

Three basic techniques have been used to describe tire flip techniques: the sumo, the backlift style, and shoulders-against-the-tire technique (21, 147). The sumo and backlift styles are somewhat similar in that the feet are placed close to the tire, but they differ on the width of the feet, somewhat similar to the sumo and conventional deadlift starting positions, respectively (147). With these two techniques, once the tire has been raised to hip or chest height, the hands are rotated so that a forward pressing action can be performed with the arms to flip the tire (147). The shoulders-against-the-tire technique has a similar starting position to a rugby scrum or prowler push, with the athlete's face almost touching the tire, trunk horizontal, and the feet positioned behind the body so that the shins are horizontal, with the knees positioned slightly in front of the hips and flexed to approximately 90 degrees (21).

Limited research has investigated tire flipping techniques (73, 162). Keogh and colleagues (73) primarily quantified the phases and temporal characteristics most related to successful (fast) completion of 2 sets of 6 repetitions of a 232-kilogram (511 lb) tire flip, separated by 3 minutes' rest. The phases of the movement were broken down into a first pull, second pull, transition, and push, somewhat similar to the clean and jerk. Results suggested that minimizing time during the second pull and push phases was most associated with faster flips, suggesting that the tire flip may be used as an alternative to exercises such as clean and jerk to develop total body power. To optimize this development of total body power, the athlete should emphasize a powerful triple extension of the lower extremities to generate sufficient momentum for the tire to continue to rotate upward and forward before the push phase. This will also require the selection of appropriately sized tires that allow this rapid triple extension and transition to the push phase so that each repetition is performed relatively quickly (~3-4 s).

Tire Flipping (Shoulders-Against-the-Tire Technique): Common Technical Flaws and Corrections

- *Flaw: Placing the feet too close to the tire when initiating the movement.*
When this occurs, athletes often have to round their back and position their knees close to their chest in order to initiate the movement.
Correction: Have athletes move their feet away from the tire, and instruct them to brace their spine and use a forceful triple extension of the lower extremities that raises their chest and tire off the ground.
- *Flaw: Hips rise faster than the shoulders during the initial pushing motion.*
This flaw is very similar to what can be seen during traditional deadlifting with incorrect technique.
Correction: Instruct athletes to keep their hips low and drive the tire forward rather than lifting it.
- *Flaw: A lifting motion is used instead of a pushing motion.*

With heavier tires, this motion reduces the lifting speed; the tire will lose momentum as it is elevated to hip height, forcing the athlete to “muscle” the tire over. This position places substantially greater load on the biceps and the risk of biceps injury, and the tire is also more likely to fall backward onto the athlete.

Correction: The athlete should be encouraged to drive the tire forward and move forward with the tire as it is elevated. One cue is to strike the tire with the quadriceps at hip height and continue a forward movement pattern.

Adapted by permission from J.B. Bullock and D.M.M. Aipa, “Coaching Considerations for the Tire Flip,” *Strength and Conditioning Journal* 32, no. 5 (2010): 76-77.

Log Lifting

One of the classic exercises that is a part of strongman training is the **log lift**, which is essentially a version of the clean and overhead press. Other traditional lifting movements can also be performed with logs, such as cleans, presses, jerks, rows, squats, deadlifts, and lunges (59, 108, 156). **Logs** are typically made from metal and designed to have weight added to them while offering a midrange grip support to accommodate a neutral grip position (108). Weight is typically added with the use of traditional plates, which eliminates much of the need for having a variety of logs (59).

The biomechanics of the log lift have been examined in two studies (110, 155). Using six strongman athletes, Winwood and colleagues (155) compared the log lift and clean and jerk, using loads of 70% 1RM clean and jerk. Both exercises involved triple extension of lower extremities and the production of high levels of vertical force. The log lift produced significantly greater impulse as well as hip and trunk range of motion but significantly less vertical velocity and power than the clean and jerk. Such differences probably reflected the greater challenges in rapidly moving the body and the log due to its greater diameter than the barbell. Such a view is supported by Renals and colleagues (110), whereby a thinner log (250 mm [1.0 in.] diameter) allowed 10 strongman athletes to produce significantly greater power, velocity, and force than a thicker log (316 mm [1.2 in.] diameter).

Weighted Carries

Strongman competitions commonly involve a range of **weighted carries** including the farmer's walk, yoke walk, or sandbag or keg carries, with regular gym programs potentially also involving Zercher carries (using a barbell held across the elbows on the upper abdominals) or unilateral carries in which the load is held overhead or by one hip (i.e., suitcase carry). A primary advantage of these exercises is that they often involve horizontal motion and unilateral ground contact and require coordinated activity of many muscles of the body. Weighted carries may therefore develop total body and trunk stability, grip strength (for farmer's walks) as well as the ability to produce high levels of anterior-posterior and vertical impulse to initiate propulsion and to be able to absorb these forces on impact at heel strike (163). These exercises therefore may mimic performance in a range of occupational and sporting demands (62, 87) or create substantial metabolic demands that can be exploited in a metabolic or anaerobic conditioning circuit (55, 163). The following exercises are common weighted carries.

- **Farmer's walk:** The farmer's walk requires the athlete to hold a load at the sides in each hand while walking forward (87, 147). The farmer's walk can be performed with static loads (e.g., farmer's bar handles or a heavy dumbbell) or variable loads (e.g., water-filled objects) (147). Regardless of the type of load used, the farmer's walk offers a unique activation pattern of the core (87), although research in the scientific literature is very limited in supporting the use of this exercise as a strength and conditioning tool. Two biomechanical studies of the farmer's walk have been performed (71, 154). Winwood and colleagues (154) demonstrated significantly greater vertical and anterior forces in the liftoff phase of the farmer's walk than the deadlift, as well as significantly larger forces and shorter ground contact times and swing times in the farmer's walk than unloaded

walking. Keogh and colleagues (71) identified that compared to slower trials, the fastest trials had significantly greater stride length and rate and reduced ground contact time.

- **Yoke walk:** The yoke walk is another common strongman event whereby a heavy frame is carried on the shoulders of the athlete over a set distance. As McGill and colleagues (87) observed significantly greater spinal compression forces during the performance of the yoke walk when compared to six other strongman events, it is likely that this exercise has the potential to challenge and improve core stability. However, it must be performed with appropriate loads and technique to reduce the risk of injury, especially since Winwood and colleagues (154) reported that 213 strongman athletes ranked this exercise as the second most dangerous strongman lift in terms of injury risk. Similar to the farmer's walk (71), faster yoke walks were characterized by increased stride length and stride rate, lower limb range of motion, and reduced ground contact time (61). Based on the impulse–momentum relationship, it would appear that weighted carries (e.g., farmer's and yoke walks) require the production of high levels of anterior–posterior and vertical forces over short contact times, perhaps suggesting some potential to improve horizontal motion such as acceleration sprinting or initiating body contact in tackling or blocking opponents in contact or combat sports.
- **Sandbag carries:** Limited research has examined frontal carries including sandbags and water bags in athletic populations (22, 60). Calatayud and colleagues (22) compared the muscle activity (anterior deltoid, external oblique, lumbar erector spinae, and gluteus medius) associated with the performance of 20-kilogram (44 lb) clean and jerks with either a barbell, sandbag, or water bag in 21 male university students. While no significant differences were observed between the muscle activity when lifting a barbell or sandbag,

the water bag version resulted in significantly greater external oblique, lumbar erector spinae, and gluteus medius activation.

- **Unilateral carries:** Unilateral carries with the load positioned overhead (e.g., kettlebell or dumbbell) or at the hip via a kettlebell, dumbbell, or farmer's handle (suitcase carry) impose a unique frontal plane challenge to the athlete (87). These carries may be advanced total body exercises that will improve lateral trunk flexor muscles (e.g., obliques and quadratus lumborum) and scapular control, thereby allowing better lateral stabilization of the spine than more traditional exercises such as dumbbell side bends or side planks.

Technique Tips for Weighted Carries

Irrespective of where the load is positioned on the body, the athlete should look to move in a similar way to normal walking. For example, during the carrying component of the exercise, the athlete should retain a predominantly neutral and vertical spinal position with no substantial hyperextension, flexion, or lateral spinal flexion. The athlete should use loads that challenge his or her desired walking speed, step length, step rate, and ground contact time but are still within his or her ability to maintain core and total body stability. In order to achieve this, the athlete will need to concentrate on keeping the loads in the same relative position to his or her center of mass throughout the carry. To achieve this, the athlete will need to stiffen the spine in response to the likely perturbations these awkward loads impose on the system during the walking phase. This may require some breath holding, which can be challenging due to the high spinal compressive forces (especially observed in the yoke walk) or if the load is positioned on the chest as in Zercher, sandbag, and keg carries, because such loads can further restrict typical respiration patterns.

Kettlebell Training

Recently, strength and conditioning professionals have become interested in the use of **kettlebell** exercises (11, 23, 83). While seemingly a new phenomenon, the use of kettlebells dates back hundreds of years to when they were a popular training method in various Eastern Bloc countries (57). The word *kettlebell* comes from the Russian word *girya*, which refers to a cast iron cannonball with a

handle on it (28). In Western literature the term *kettlebell* is used to refer to a weighted implement consisting of a ball with a handle, which is probably better termed a *kettleball* (28).

Along with the increasing interest in the use of kettlebell training has been an increase in the amount of scientific inquiry into the efficacy of using these implements with athletes and general populations. The majority of the scientific evidence supporting the use of kettlebells highlights their potential usefulness as a tool for general physical or fitness development (38, 65). In these studies, the most common exercise employed is the kettlebell swing, which can be performed with either one or two hands (65, 87, 137). While kettlebell swings appear to have a positive impact on cardiovascular fitness, it is important to note that this activity does not offer the same level of cardiovascular benefit as treadmill running or more traditional aerobic exercise (65).

Currently, limited research explores the efficacy of kettlebell training as a strength development tool. When used with clinical and recreationally trained populations, kettlebell training has been reported to increase muscular strength levels compared to no training (68, 103). Additionally, when a battery of kettlebell exercises (i.e., swings, goblet squats, accelerated swings) is performed across a six-week training intervention, there appears to be an increase in muscular strength and vertical jump performance in recreationally trained men (103). However, these strength gains are significantly lower than those typically seen with traditional weightlifting-based training methods. Specifically, Otto and colleagues (103) reported that six weeks of weightlifting training resulted in a 4% increase in vertical jump performance, while kettlebell swings increased jumping performance by only 0.8%. Additionally, back squat strength was increased by 13.6% after six weeks of traditional weightlifting training, while the kettlebell training resulted in only a 4.5% increase (103).

Based on the contemporary body of scientific knowledge, it appears that kettlebells are probably best used as general

preparation exercise and that more traditional training methods such as weightlifting are more effective for developing maximal strength and jumping performance capacity. However, further scientific research is needed to develop a better understanding of the role that kettlebells may be able to play in the development of athletes.

Types of Kettlebells

When working with kettlebells, one must consider the type of kettlebell, of which there are two: the cast iron ([figure 17.9](#)) and the sport kettlebell ([figure 17.10](#)) (28). The cast iron, or fitness, kettlebells are cast from iron and range in size depending on their weight. The sport or competition kettlebells are made from steel and have universal design and measurements. Specifically, the size of the kettlebell does not change regardless of the weight, and the various weights are indicated by different colors (28). Besides the structural differences in the size and dimensions of the cast iron and sport kettlebell, the other difference is that the cast iron kettlebell is less expensive and probably more prevalent in strength and conditioning facilities.

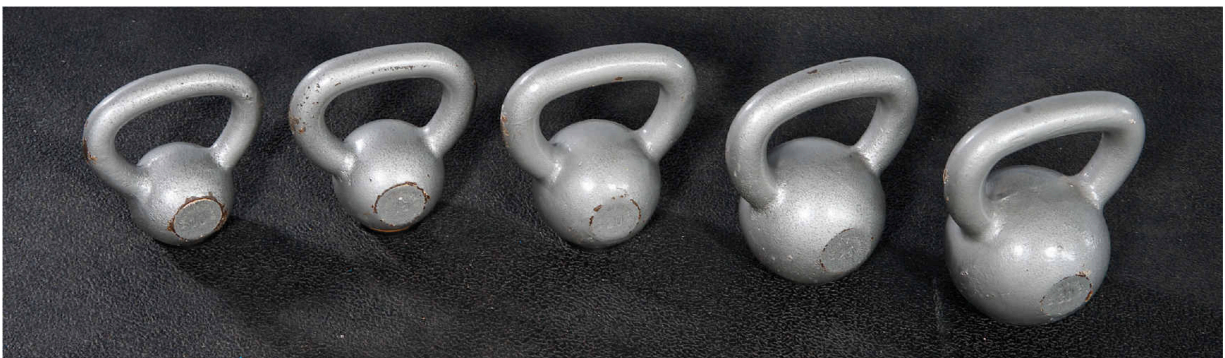


FIGURE 17.9 Cast iron class kettlebells.



FIGURE 17.10 Competition kettlebells.

Competition Kettlebell Dimensions

Following are the dimensions of competition kettlebells (28):

- *Height:* 228 mm (9.0 in.)
- *Diameter:* 210 mm (8.3 in.)
- *Handle diameter:* 35 mm (1.4 in.)

Considerations for Selecting Kettlebells

The first major consideration in the selection of a kettlebell is the type of load provided. The two basic types of kettlebells are the fixed and the adjustable (28). With fixed-loaded kettlebells, such as cast iron or competition kettlebells, the load stays constant; thus, a set of kettlebells, which range across several loads, is required in order to provide training variety. Adjustable kettlebells are either plate loaded or shot loaded. In actuality, a plate-loaded kettlebell is simply a handle that is attached to weight plates and is not really a kettlebell even though it is classified as one (28). In contrast, a shot-loaded kettlebell is a hollow version of the more traditional kettlebell. Historically, these kettlebells were filled with sand, water, lead, and even mercury (28). If this type of kettlebell is only partially filled, the shot moves around in the ball, causing an increased training stressor. While popular in the early 20th century, especially with circus strongmen such as the famous Arthur Saxon and Eugen Sandow, this type of kettlebell is not popular in contemporary strength and conditioning.

The second consideration relates to the handle, because it is the major interface between the athlete and the kettlebell. With cast iron kettlebells, the handle diameter changes slightly as the weight increases. For example, a 20-kilogram (44 lb) or heavier kettlebell has a handle between 33 (1.3 in.) and 35 (1.4 in.) millimeters; a smaller kettlebell may have a smaller handle diameter. Additionally, to facilitate the types of exercises that use kettlebells, the spacing between the handle and the top of the ball is standardized in good kettlebells. Typically, the space between the bottom of the handle and the top of the ball is 55 millimeters (2.2 in.), and the length of the handle is 186 millimeters (7.3 in.) (28). With regard to handle surface, some kettlebells come with smooth painted handles; others have polished steel handles, which have no paint and are basically bare metal. The polished steel handle tends to allow for a better grip because it holds chalk better and does not get as slippery as painted or polished handles do when athletes sweat on them (28).

Landmine Exercises

The **landmine** resistance training device is a useful device for performing loaded rotational resistance training exercises (4). The performance of this exercise involves placing the end of a standard Olympic barbell (i.e., 20 kg [44 lb]) into a metal cylinder that is attached to a standard squat rack or stand-alone base attachment through a universal joint (4). With this set-up, the barbell is used as a lever and is rotated around the fixed end, allowing for loaded rotational movements to be performed (4). Alternatively, if a purpose-built receptacle is not available, one end of the barbell can be placed in contact with the floor at the corner of a wall and a similar movement pattern can be performed (148). Regardless of the how the barbell is anchored, additional resistance can be added to the free end of the barbell, thus allowing for the application of variable and multiplanar resistance during rotational movements (26, 80, 167).

Additionally, the landmine device can be used to perform a variety of exercises including the landmine row (80), Romanian deadlift (RDL) (85), squat (26, 114), press (135), and squat to press (69). While these exercises have become commonly implemented as part of the resistance training practices of many athletes (4), there is a paucity of scientific research exploring the actual physiological impact and efficacy of these exercises. As such, significantly more research is warranted to understand the unique training stimulus provided by landmine exercises.

Sled Training

In addition to the training studies incorporating sleds as a part of the conditioning program, several studies have examined the acute responses to sled exercise (72, 150, 152, 160).

Keogh and colleagues (72) sought to describe the spatiotemporal characteristics of the heavy sled pull, using six resistance-trained subjects with experience in the heavy sled pull. Participants performed 3 25-meter (27 yd) sets of sled pulls with 171.2 kilograms (377 lb), with 3 minutes of rest between sets. Several significant differences were observed between the acceleration (the start through 5 m [5 yd]) and maximum velocity (20-25 m [22-27 yd]) phases, with average velocity, step length, swing time, and a range of segment and joint angles being significantly greater in the maximum velocity than acceleration phase. The fastest trials also had significantly greater step length, swing times, and trunk and knee angles at certain positions than slower trials. These acute responses would potentially underlie some of the mechanisms contributing to the chronic improvements in sprinting performance seen with heavy sled pulls (96).

West (150) examined the acute effects of five sets of 2 × 20-meter (21 yd) reverse sled pulls using loads of 75% of body mass with 11 resistance-trained men. Participants were retested immediately, 15 minutes, 1 hour, 3 hours, and 24 hours after completing the sled pulls. Peak countermovement jump power was

significantly decreased and blood lactate significantly increased (peaking at 12.4 mmol/L immediately after) for a period of 1 hour, with no significant changes in creatine kinase at any time point. Testosterone and cortisol tended to increase at 15 minutes after (significant only for testosterone), with no significant changes in the testosterone-to-cortisol ratio observed. Such results suggest that concentrated sled pull exercises can produce profound metabolic fatigue and short-term hormonal augmentation with no muscle damage. The relatively quick recovery and lack of muscle damage (perhaps due to the concentric-dominant nature of this exercise) may indicate that lower and upper body sled exercises could be a key component of a resistance training program when athletes' recovery abilities are particularly challenged, such as when they are performing large volumes of concurrent training or competing several times per week.

Winwood and colleagues (160) and Williams and colleagues (152) have examined the potential of moderate to heavy sled pulls to produce a postactivation potentiation enhancement (PAPE) sprinting effect in adult semi-elite rugby (75%-150% body mass) and adolescent high school soccer players (loads producing a velocity decline of 40%-50% that equated to 66%-70% body mass), respectively. Some significant PAPE effects were observed that resulted in significant reductions in 15-meter (16 yd) sprint times by Williams and colleagues (152) and Winwood and colleagues (160). However, the significant PAPE benefit for Winwood and colleagues (160) was only observed using 75% body mass loads and at 12 minutes after completing the sled pulls. The studies therefore suggest that sled pulls with loads between 66% and 75% of body mass have the potential to produce 15-meter sprint PAPE effects in a variety of running athletes.

Unilateral Training

When employing training methods, one can use unilateral or bilateral training interventions. These types of training methods can be performed with the upper body or lower body, depending on the targeted outcomes. Common lower body unilateral training exercises include lunges, step-ups, and the single-leg squat, which is sometimes referred to as the *Bulgarian split squat*. This exercise isolates one leg and is typically used in the preparation of athletes from numerous sports. Typically, these types of exercises are integrated into training programs with varying degrees of emphasis (86) in an attempt to reduce **bilateral asymmetries** (75) or as a rehabilitation tool (37). They are often used to reduce a **bilateral deficit**, where asymmetries in force production are observed between unilateral and bilateral movements (67). It has also been demonstrated that bilateral movements exhibit a **bilateral facilitation** in which voluntary activation of the agonist muscle group increases (12, 117). Trained or stronger individuals tend to exhibit a bilateral facilitation, while untrained, injured, or weaker athletes exhibit a bilateral deficit (14). Therefore, based on the bilateral facilitation response, trained individuals should not use unilateral methods as the primary strategy to develop strength. In contrast, unilateral training methods may be useful for strength development with untrained, weaker, and injured individuals (14).

Trained or stronger individuals exhibit a bilateral facilitation during bilateral exercises, while untrained, injured, or weaker individuals exhibit a bilateral deficit.

Conclusion

In designing strength and conditioning interventions, one can use a variety of methods to apply overload. The use of alternative methods and nontraditional implements is increasing in popularity. When choosing to implement these methods, strength and conditioning professionals should consider the benefits and weaknesses of these types of training interventions. Additionally, the level of the athlete dictates whether use of these methods is warranted. For example, a novice athlete or untrained individual may derive a large benefit from doing bodyweight or core stability exercises. Conversely, the trained or elite athlete may experience greater gains with traditional ground-based free weight exercises. With the advanced athlete, the use of variable resistance may also allow for a greater training stimulus to be applied to traditional training methods. If choosing to employ alternative methods or nontraditional element-based training methods, strength and conditioning professionals must always ensure that they teach proper exercise techniques as well as constantly monitor their athletes to ensure that a safe training environment is maintained.

ALTERNATIVE MODES AND NONTRADITIONAL EXERCISES

BODYWEIGHT EXERCISES

- 1 Front Plank [498](#)
- 2 Side Plank [498](#)
- 3 Suspension Trainer Chest Press [499](#)
- 4 Suspension Trainer Inverted Row [500](#)
- 5 Suspension Trainer Y Pull [501](#)

CORE STABILITY AND BALANCE TRAINING EXERCISES

- 6 Stability Ball Rollout [502](#)
- 7 Stability Ball Pike [503](#)
- 8 Stability Ball Jackknife [504](#)

STRONGMAN EXERCISES

- 9 Tire Flip [505](#)
- 10 Log Clean and Press [506](#)
- 11 Yoke Walk [508](#)
- 12 Sandbag Carry [509](#)
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OTHER ALTERNATIVE EXERCISES

- 16 Back Squat With Bands [515](#)
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- 18 Landmine Rotation [517](#)
- 19 Single-Arm Landmine Row [518](#)
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- 24 One-Arm Dumbbell Snatch [523](#)

Bodyweight Exercises

1 FRONT PLANK

Starting Position

- Kneel in a prone, quadruped position on the floor. The feet should be hip-width apart or slightly closer, with the palms of the hands flat on the floor shoulder-width apart and the elbows pointing backward.
- Drop the elbows to the floor and position them directly under the shoulders.
- Move the feet back one at a time to allow the hips and knees to extend so the abdomen and the front of the hips and legs rest on the floor.

Ending Position

- Elevate the hips so the ankles, knees, hips, shoulders, and head are in a straight line.
- Isometrically hold the torso in a rigid position with the elbows under the shoulders and the head in a neutral position.

MAJOR MUSCLES INVOLVED

rectus abdominis, internal obliques, external obliques, erector spinae



Starting position



Ending position

2

SIDE PLANK

Starting Position

- Lie on the floor on the left side.
- Position the left elbow underneath the left shoulder with the left forearm perpendicular to the body.
- Stack the right foot on top of the left foot with the right leg evenly on top of the left leg, or place the right foot on the floor immediately in front of the left foot. Also, stack the right arm evenly on top of the right side of the torso.
- Position the head in a neutral position with the eyes focused forward.

Ending Position

- Elevate the hips off the floor so the left ankle, knee, hip, and shoulder are in a straight line.
- Isometrically hold the torso in a rigid position with the left elbow under the left shoulder and the head in a neutral position.
- Switch and repeat the procedure for the right side.

MAJOR MUSCLES INVOLVED

internal obliques, external obliques



Starting position



Ending position

3 SUSPENSION TRAINER CHEST PRESS

Starting Position

- Assume a position away from the anchor point while grabbing the handles of the suspension trainer in each hand.
- With the elbows extended, move the hands and feet shoulder-width apart while placing the body into a straight line, or plank position.
- Brace the trunk and slowly step back until there is tension on the straps and the body is in an inclined position.

Movement

- Initiate the movement by flexing the elbows and lowering the chest, similar to a push-up.

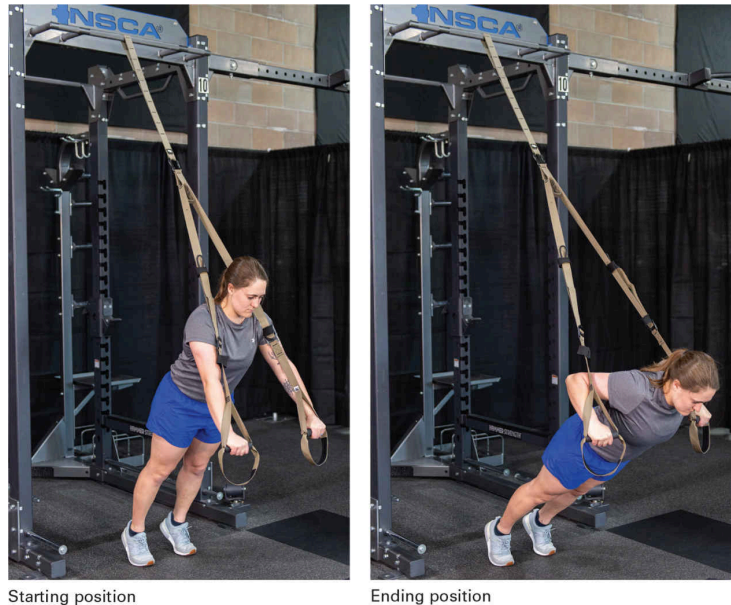
Ending Position

- At the end of the movement the chest should be between the suspension trainer's handles.

- Extend the elbows to return to the starting position.

MAJOR MUSCLES INVOLVED

pectoralis major, anterior deltoid, triceps brachii, rectus abdominis, internal obliques, external obliques, erector spinae



4 SUSPENSION TRAINER INVERTED ROW

Starting Position

- Face the anchor point while grabbing the handles of the suspension trainer with the palms of the hands facing each other or upward.
- While maintaining a neutral body position, lean back while simultaneously allowing the arms to extend in front of the body until they are at or above shoulder height.

Movement

- Pull the torso toward the hands by flexing at the elbows and extending the shoulders. Throughout the movement, maintain a

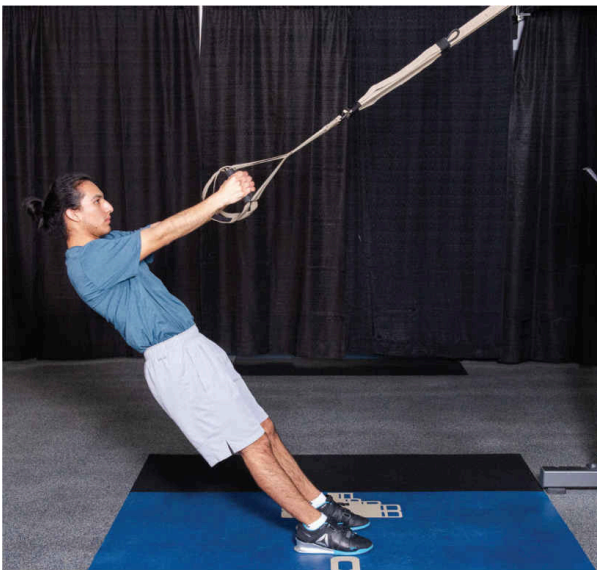
straight body posture.

Ending Position

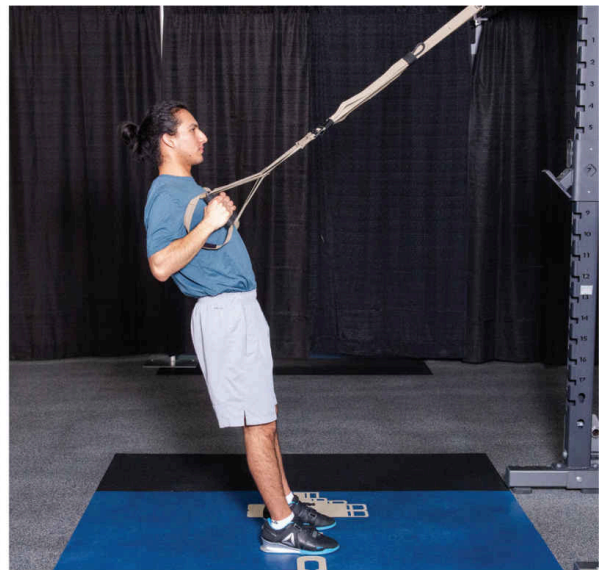
- The elbows are flexed, and the hands are next to the chest.
- Extend the elbows, and lower the torso back to the starting position.

MAJOR MUSCLES INVOLVED

latissimus dorsi, rhomboids, biceps



Starting position



Ending position

5

SUSPENSION TRAINER Y PULL

Starting Position

- Face the anchor point while grabbing the handles of the suspension trainer with the palms of the hands facing each other.
- While maintaining a neutral body position, lean back while simultaneously allowing the arms to extend in front of the body until they are at or above shoulder height.

Movement

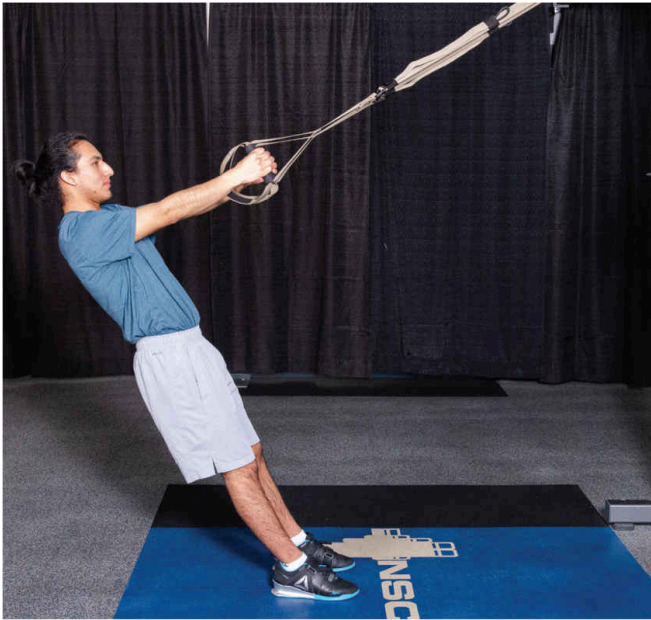
- Without flexing the elbows, flex the shoulders while pushing the hands overhead in a diagonal movement pattern to form a Y with the arms and body.
- Throughout the movement, maintain a straight body posture.

Ending Position

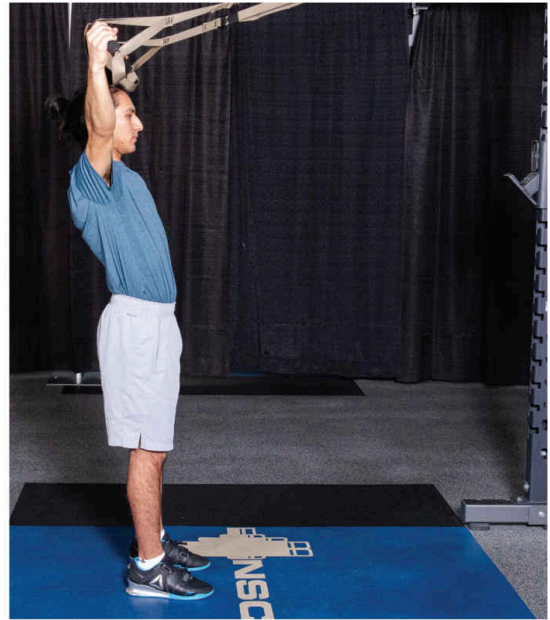
- At the top of the movement the arms and body should form a Y.
- After pausing in this position, slowly lower the body back to the starting position.

MAJOR MUSCLES INVOLVED

latissimus dorsi, rhomboids, trapezius, posterior deltoid



Starting position



Ending position

Core Stability and Balance Training Exercises

6

STABILITY BALL ROLLOUT

Starting Position

- Kneel facing the stability ball with the upper body in an upright position, the elbows fully extended, and the hands touching the ball.
- While keeping the hands on the ball, create a 90-degree angle at the knees and ankles with the knees, hips, and shoulders in a near-vertical plane.

Ending Position

- Keep the knees and toes on the floor; elbows fully extended; arms parallel to each other; and knees, hips, and shoulders in a straight line.
- Extend the knees and flex the shoulders to roll the ball forward and the arms across the top of the ball until it comes very close to the face.
- Isometrically hold the torso in a rigid position; do not let the hips sag toward the floor.

MAJOR MUSCLES INVOLVED

rectus abdominis, iliopsoas



Starting position



Ending position

7

STABILITY BALL PIKE

Starting Position

- Kneel down in front of a stability ball.

- Position the body facing away from the ball.
- In a quadruped position with the hands underneath the shoulders and the knees extended, place both feet on the stability ball.
- Isometrically hold the torso in a rigid position with the elbows fully extended and the head in a neutral position.
- All repetitions begin from this position.

Upward Movement Phase

- Keeping the knees and elbows fully extended and the torso in a rigid position, begin the exercise by flexing the hips to roll the ball toward the chest.
- Continue lifting the hips until the toes are on top of the ball and the hips are directly over the shoulders.
- Keep the head in a neutral position.

Downward Movement Phase

- Return to the starting position by allowing the hips to extend under control.
- Keep the knees and elbows fully extended and the torso held rigid.

MAJOR MUSCLES INVOLVED

rectus abdominis, iliopsoas



Starting position



Upward and downward movements

8

STABILITY BALL JACKKNIFE

Starting Position

- Kneel down in front of a stability ball.
- Position the body facing away from the ball.
- In a quadruped position with the hands underneath the shoulders and the knees extended, place both feet on the stability ball.
- Isometrically hold the torso in a rigid position with the elbows fully extended and the head in a neutral position.
- All repetitions begin from this position.

Forward Movement Phase

- Keeping the knees and elbows fully extended and the torso in a rigid position, begin the exercise by raising the hips slightly and flexing the hips and knees to roll the ball toward the chest.
- Continue rolling the ball forward until the hips and knees are fully flexed.
- Keep the shoulders over the hands and the head in a neutral position.

Backward Movement Phase

- Return to the starting position by allowing the hips and knees to extend under control.
- Keep the elbows fully extended and the torso held rigid.

MAJOR MUSCLES INVOLVED

rectus abdominis, iliopsoas



Starting position



Forward and backward movements

Strongman Exercises

9

TIRE FLIP

Starting Position

- Begin facing the tire with the feet hip- to shoulder-width apart.
- Squat down and lean into the tire so that the chin and the anterior deltoids are resting on the tire and the feet are positioned back far enough to maintain a neutral spine position.
- Keeping the arms outside the knees and the elbows fully extended, grasp the tire with a supinated grip.
- All repetitions begin from this position.

Upward Movement Phase

- Begin by extending the knees and hips and plantar flexing the ankles while pushing the tire forward.
- Move forward explosively toward the tire by taking two or three steps.
- Continue this movement until the body is aligned in a 45-degree angle to the tire.
- Consider flexing the hip and knee of one leg fully to drive the knee upward and toward the tire.
- Immediately after striking the tire with the thigh (if done), reorient the hands into a pronated grip.

- Lean forward and take small steps as required while forcefully extending the arms to push the tire over in front of the body.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, soleus, gastrocnemius, deltoids, triceps brachii, trapezius



Starting position



Triple extension



Strike position



Push position



Walk forward

10 LOG CLEAN AND PRESS

This exercise comes from strongman training and consists of quickly and forcefully picking the log up from the ground to the front of the shoulders and then pressing it overhead. The log clean consists of three phases but occurs in two motions, with a pause of the log at

the shoulders. The log can either be cleaned and pressed each repetition or cleaned once and pressed for multiple repetitions.

Starting Position

- Stand with the feet placed between hip- and shoulder-width apart with the toes pointed forward or slightly outward.
- Squat down with the hips lower than the shoulders, and grasp the log with a closed, neutral grip.
- Place the hands on the log handles with the arms slightly outside of the knees and the elbows fully extended.
- Place the feet flat on the floor, and position the log in front of the shins and over the balls of the feet.
- All repetitions begin with the back in a neutral spine position, scapulae depressed and retracted, head in line with the vertebral column, and shoulders over or slightly in front of the log.

Upward Movement Phase: First Pull

- Lift the log off the ground by extending the hips and knees.
- Keep the torso-to-floor angle constant; do not let the hips rise before the shoulders.
- Keep the elbows extended and the shoulders over or slightly ahead of the log.

Upward Movement Phase: Transition

- As the log passes the knees, pull the log into the body by dropping into a full squat position while ensuring the elbows are about shoulder height and positioned slightly behind or above the hands.
- Briefly pause in this position with the log on the thighs.

Upward Movement Phase: Second Pull

- Forcefully jump upward by extending the hips and knees and plantar flexing the ankles.

- Ensure the log is kept close to the body throughout this phase, whereby it essentially rolls up the upper body toward the shoulders.

Upward Movement Phase: Catch

- After the lower body has fully extended, pull the body under the log and rotate the arms around under the log, similar to the catch in the power clean.
- Simultaneously flex the hips and knees to a quarter- or half-squat position.
- Rack the log across the front of the clavicles and anterior deltoids.

Preparation Phase: Dip

- Flex the hips and knees at a slow to moderate speed to move the log in a straight path downward.
- Continue the dip to a depth not to exceed a quarter squat.
- Keep the feet flat on the floor, the torso erect, and the upper arms directly under the log.

Upward Movement Phase: Drive

- Immediately on reaching the lowest position of the dip, reverse the movement by forcefully and quickly extending the hips and knees and then the elbows to move the log overhead.

Downward Movement Phase

- Lower the log by gradually reducing the muscular tension of the arms to allow a controlled descent of the log to the shoulders.
- Simultaneously flex the hips and knees to cushion the impact of the log on the shoulders, and then return the log to the floor. If sufficient drop pads are available, it is possible to drop the log from overhead to the floor. In order to do this safely, push the bar slightly forward from the finished overhead position while retaining some grip on the handles for a part of the descent.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris, deltoids, triceps brachii, trapezius



Starting position



End of first pull and beginning of transition



End of transition and beginning of second pull



End of second pull and beginning of catch



Catch and beginning of dip



End of dip and beginning of drive



End of drive (final overhead position)

11 **YOKE WALK**

Starting Position

- Adjust the yoke to a height that would provide sufficient clearance to walk safely without tripping over it (perhaps 4-6 in. [10-15 cm] of ground clearance).

- Assume a position under the yoke with the bar positioned on top of the upper trapezius muscles, with the feet positioned underneath the hips and the rest of the body symmetrically positioned under the frame. This should look something similar to the starting position of a partial barbell back squat, except that the hands are positioned on the vertical bars around lower chest height.

Liftoff and Walking Phase

- After strongly bracing the spine, forcibly push the feet into the ground while pushing the elbows and hands forward into the vertical bars to better stabilize the yoke.
- While continuing to brace the spine and push the elbows and hands forward, adopt a walking movement pattern in which the spine remains vertical; initially focus on performing a series of short and relatively fast steps.
- While accelerating, increase the length of steps and further reduce ground contact times.

Ending Position

- After completing the set distance, flex the hips, knees, and ankles to lower the frame carefully to the ground.
- Maintain a strong braced spinal position to minimize the risk of impact injury.

MAJOR MUSCLES INVOLVED

gastrocnemius, soleus, gluteus maximus, rectus femoris, gluteus medius, erector spinae, internal and external obliques

12 SANDBAG CARRY

Starting Position

- Adopt a position just behind the center of the sandbag, similar to the starting position of a barbell deadlift.